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## MEDICINE CONCENTRATIONS

SIMULATE DIFFUSION OF MEDICINE INTO BLOODSTREAM  
AND FILTRATION BY KIDNEYS.

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JIM COEN  
IRELAND

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# Summary

The purpose of this study is to investigate the flow of medicine through the body. Over-prescribing a drug can lead to adverse reactions. In addition, many medicines have side effects which may be harmful at high dose levels. Consequently, it is critical for doctors to know the amount of drug that the patient in fact is receiving.

In order for a model go be an accurate representation of medicine flow, it needs to take account of two synergistic processes: diffusion from GI tract to bloodstream and elimination from bloodstream into the kidneys. The model must take account of these processes in simultaneous operation.

Euler's Method [3] is a means to approximate the flow of medicine through the body. The first stage of modelling analyses transfer into and transfer out of the bloodstream separately. The second stage combines these effects. Finally, a full system model models change in the GI tract and in the bloodstream simultaneously.

This study arrives at appropriate values for the diffusion rates. It also answers questions concerning medicine absorbed by the body and the effect of dose amount.

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# List of variables

|                  |                                                                                          |
|------------------|------------------------------------------------------------------------------------------|
| $M_{GI}(t)$ :    | quantity medicine in GI over time [mg]                                                   |
| $V_{GI}$ :       | volume of the contents in the GI tract [litres]                                          |
| $M_{blood}(t)$ : | quantity medicine in blood over time [mg]                                                |
| $V_{blood}$ :    | blood volume [litres]                                                                    |
| $C_{GI}$ :       | Concentration of drug in the GI tract [mg / litre]                                       |
| $d_1$ :          | rate of diffusion from GI to blood [ 1/h ]                                               |
| $d_2$ :          | rate of diffusion from blood to kidneys [ 1/h ]                                          |
| $t$ :            | time [hours]                                                                             |
| $D_a$ :          | Administered daily dose, amount of medicine taken orally, assuming one dose per day [mg] |
| $D_e$ :          | Effective daily dose, amount of medicine absorbed in the GI tract [mg]                   |
| $B$ :            | Bioavailability; the systemically available fraction of a drug                           |
| $k_e$ :          | Elimination rate constant in the Pharmacokinetics model [5] [1/h]                        |

# Chapter 1

## Introduction

### 1.1 Problem Specification

To understand the relationship between drug absorption and elimination and how this effects the changing quantity of drug in the body. Previous studies considered rate of change in terms of medicine amount [1] or concentration of medicine [7]. It is necessary to take account of concentration gradient since this is the mechanism that drives diffusion.

#### 1.1.1 Research Questions

1. To understand how the two processes of diffusion from GI to blood and elimination through kidneys interact.
2. Doctors need to know the actual amount of medicine that a patient receives via the bloodstream for both single dose and repeated doses.
3. To understand the effect of dose amount.

### 1.2 Background

#### 1.2.1 Single Compartment Model

Koch [1] describes a single compartment model, shown in figure 1.1. This model treats the body as a homogenous unit. Equation 1.1 states that the rate of change in the quantity of medicine in the body is proportional to the quantity of medicine itself. The negative constant ( $-\alpha$ ) means that the amount of medicine decreases over time.

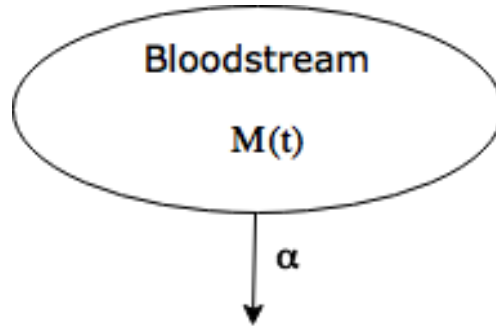


Figure 1.1: **Single Compartment Model.** The flow of medicine through the body reduces over time.

$$\begin{aligned}\frac{dM}{dt} &\propto M(t) \\ &= -\alpha M(t)\end{aligned}\tag{1.1}$$

Appendix B.1 gives the derivation of this equation in terms of initial amount of medicine. It shows that the quantity of medicine will decay exponentially after administering a single dose, given in equation 1.2.

$$M(t) = M_0 e^{-\alpha t}\tag{1.2}$$

### 1.2.2 Two Compartment Model

Koch [1] also describes a two compartment model consisting of GI tract and bloodstream. There is transfer of medicine from GI tract in to the bloodstream and transfer out from the bloodstream to kidneys. However, rate of change of medicine is in terms of medicine amount. There is no account of concentration gradient.

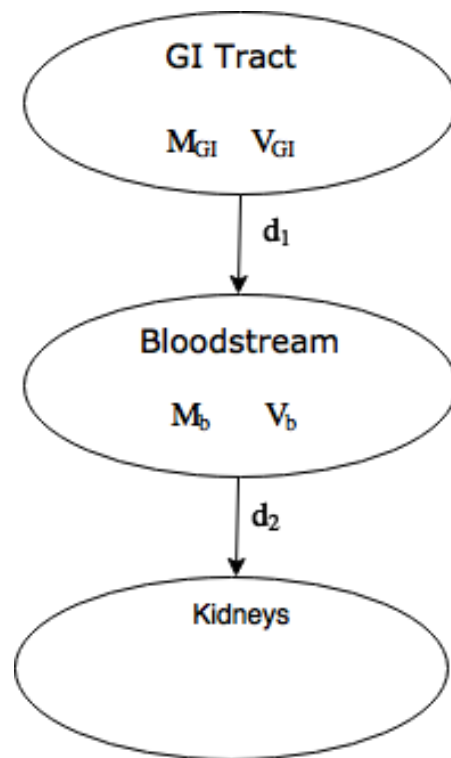


Figure 1.2: **Two Compartment Model.** Diffusion from GI tract to bloodstream and filtration through kidneys.

## Chapter 2

# Mathematical Model

This study uses the two compartment model described in 1.2.

### 2.1 Equations

The formulas describing the change in quantity of medicine in the GI tract and blood are:

$$\frac{dM_{GI}}{dt} = -d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \quad (2.1)$$

$$\frac{dM_{blood}}{dt} = d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \quad (2.2)$$

Equation 2.1 gives the rate of diffusion from the GI tract to the bloodstream. Similar to the single compartment model in section 1.2.1, the sign of the diffusion constant,  $-d_1$ , means that the rate of medicine flow from the GI tract is negative - medicine is flowing out and the amount in the GI tract ( $M_{GI}$ ) is decreasing. The flow of medicine reduces as the concentration gradient between GI tract and bloodstream decreases. So there is a levelling off similar to exponential decay. For instance, a patient takes a single dose of 400mg. Initially, the flow of medicine from GI to blood is high. The flow reduces as the difference in concentration between GI tract and bloodstream lowers. Assuming there is no filtration into the kidneys, the levels of drug in the GI tract and bloodstream will tend to equalise at 200mg in both compartments.

Equation 2.2 describes the relationship between diffusion into the bloodstream and filtration out through the kidneys. The first term in the equation describes

transfer into the bloodstream. The quantity of drug in the blood will rise steeply at first due to the concentration gradient. As the amount of drug in the blood increases, so does the amount in the GI decrease. So the rate of diffusion reduces until the amount of drug in the blood reaches the same amount as in the GI tract.

It is possible to consider the kidneys as drain or sink since the kidneys does not become saturated with medicine (blood is constantly filtered through the kidneys and the amount of drug to eliminate is decreasing). The second term in 2.2 describes transfer out from the bloodstream. It specifies that elimination into the kidneys is proportional to the concentration of medicine in the blood. As the kidneys clear more and more drug from the blood, the rate of elimination reduces gradually. Rate of flow does not rely on concentration gradient. The rate constant  $d_2$  is similar to  $\alpha$  in the single compartment model in section 1.2.1.

Taken as a whole, equation 2.2 formulates that change of medicine amount equals transfer in minus transfer out. Overall it suggests that the change in quantity of medicine over time ( $\frac{dM_b}{dt}$ ) exhibits exponential growth followed by exponential decay with a damping effect for higher quantities of drug in the blood - incoming from the GI tract leads to outgoing into the kidneys. In addition, as the amount of drug in the blood increases, so does the amount in the GI decrease, which contributes to the damping effect.

## 2.2 Stability

Evaluation of stability uses a simplified model which assumes that the ratio of medicine amount to volume of contents of GI tract is constant. This is a reasonable assumption given that both quantities reduce over time. The assumption here is that they reduce in proportion.  $C_{GI}$  denotes this constant level for the concentration of drug in the GI tract.

$$C_{GI} = \frac{M_{GI}(t)}{V_{GI}}$$

So equation 2.2 becomes:

$$\begin{aligned} \frac{dM_{blood}}{dt} &= d_1 \left( C_{GI} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \\ &= d_1 C_{GI} - \left( \frac{d_1}{V_b} + \frac{d_2}{V_b} \right) M_{blood}(t) \end{aligned} \tag{2.3}$$

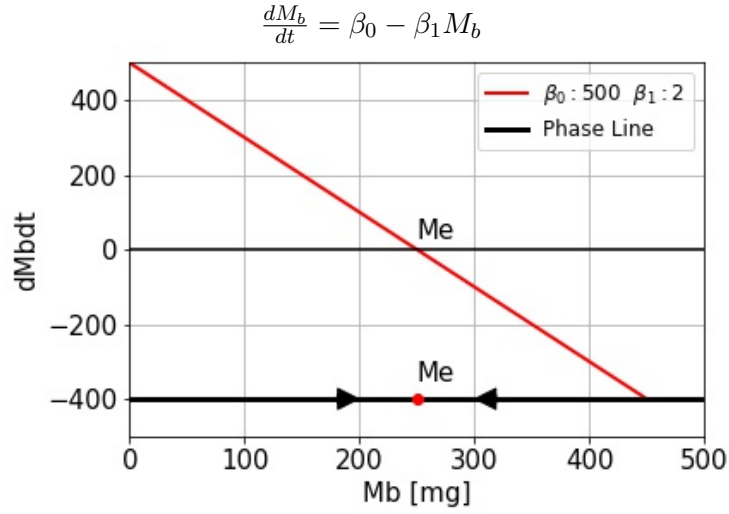


Figure 2.1: **Differential against amount of medicine.** Rate of change of medicine amount in bloodstream for values of  $M_{blood}$ .

Given that  $V_{blood}$  is also constant and that both  $d_1$  and  $d_2$  are positive, the equation 2.3 has the form:

$$\frac{dM_b}{dt} = \beta_0 - \beta_1 M_b \quad (2.4)$$

Figure 2.1 is a plot of the medicine differential at values of medicine amount given by equation 2.3. This shows that there is an equilibrium point at  $M_e$ . Superimposed on the graph is the phase line. Below  $M_e$ , the medicine flow differential is positive, so the arrow on the phase line is in the increasing direction. For values greater than  $M_e$ , the differential is negative and the arrow is in the decreasing direction. This means that, when the amount of drug drops below  $M_e$ , the positive rate of change causes  $M_{blood}$  to increase back to equilibrium. Similarly, when the drug amount is greater than  $M_e$ , the negative rate of change tends to bring that amount back towards equilibrium. Therefore, the equilibrium point is stable.

## 2.3 Evaluation of Quantities from know metrics

The analysis here uses the metrics for a typical drug in the Wikipedia article on Pharmacokinetics [5]. It assumes a single dose of drug.

$$\begin{aligned} D_a &= 500mg && \text{(Dose administered)} \\ B &= 0.8 && \text{(Bioavailability; the systemically available fraction of a drug.)} \\ D_e &= B \times D_a = 400mg \end{aligned}$$

$$\begin{aligned}
C_{max} &= 60.9 \text{ mg/L} && \text{(the effective dose entering the GI tract)} \\
t_{max} &= 3.9 \text{ h} && \text{(The peak plasma concentration of a drug after administration)} \\
V_{blood} &= 6 \text{ L} && \text{(Time to reach } C_{max} \text{).}
\end{aligned}$$

### 2.3.1 Diffusion from GI to blood ( $d_1$ )

Plasma makes up 55% of total blood volume [6]. There is 6 litres of blood in the bloodstream. So 6 litres of blood contains 3.3 litres of plasma. And the maximum amount of medicine in 3.3 litres of plasma is  $60.9 \times 3.3 = 200.97 \text{ mg}$  of medicine. Thus the maximum amount of medicine in the bloodstream is:

$$M_{max} = 200.97 \text{ mg}$$

Note that this value concurs with the analysis of equation 2.1 where amounts of medicine in GI tract and bloodstream equalise over time.

From these values, we have 200.97 mg entering the bloodstream in the time period  $t_{max}$  subsequent to administering a single drug dose. The following analysis assumes that elimination through the kidneys does not have a significant effect until after reaching  $M_{max}$ .

Comparing with the population growth model [2], 2 fish produce 42 new fish in 30 days. In 1 day, 2 fish produce 1.4 new fish and 1 fish produces 0.7 new fish. So the birth rate, the per day rate for one fish, is 0.7.

Similarly with the medicine model, in one hour, the amount of medicine that diffuses into the blood is:

$$\frac{200.97}{3.9} = 51.53 \text{ mg/h}$$

So one unit of medicine diffuses at a rate of  $1/51.53$  per hour. Consequently, the per hour rate for one mg to diffuses from GI tract to bloodstream is:  $d_1 = 0.0194 \text{ h}^{-1}$

### 2.3.2 Diffusion from blood to Kidneys ( $d_2$ )

The Pharmacokinetics article [5] gives the elimination rate constant:

$$k_e = 0.0578 \text{ h}^{-1}$$

This refers to drug concentration in plasma. Scaling by a factor of  $\frac{3.3}{6}$  gives the comparative value for blood:

$$k_{eb} = 0.03179 \text{ h}^{-1}$$

From the analysis above, estimates for the diffusion rates are:

$d_1 = 0.0194$  and

$d_2 = 0.03179$

This means that elimination of drug through the kidneys occurs faster than infusion from the GI tract into bloodstream. This indicates that the analysis above need re-evaluation

In comparison, the Alexandria study of drug diffusion [7], gives the comparable values of  $d_1$  at 0.977 while  $d_2$  ranges from 0.2 to 0.7  $h^{-1}$ . Here again, researchers were modelling changes in concentration of drug.

### 2.3.3 Volume of Contents of GI Tract

There is great variability in the volume of the contents of the GI tract [8]. This means that the greatest uncertainty in the model is in this value. Lectures on surgery given by Ireland [11] give a value of 4 litres. So the concentration of medicine in the GI tract is:

$$C_{GI} = \frac{M_{GI}}{V_{GI}} = \frac{D_e}{4} = 100 \text{ mg/litre}$$

As stated in section 2.2, the analysis for the simplified model assumes that  $C_{GI}$  is constant.

## Chapter 3

# Simplified Model

In the following models, an intake of a 500mg of medicine only reaches an effective dose level of 400mg due to bioavailability, as outlined in section 2.3.1

### 3.1 Single Compartment Model

Recall that the mathematical model of the single compartment model (SCM) is:

$$dM/dt = -\alpha M(t)$$

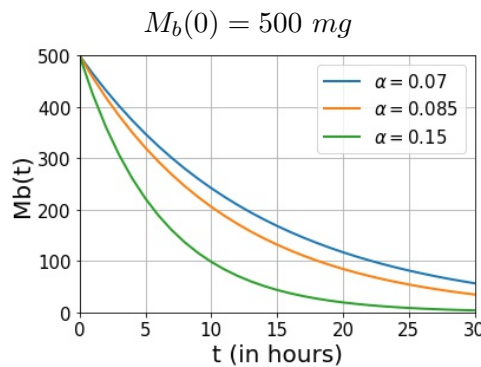


Figure 3.1: **Euler's approximation for SCM.** Euler's method approximations for SCM at values of  $\alpha$  from 0.15 to 0.07

Euler's approximation for the medicine change differential confirm the derivation given in appendix B.1, which concluded that  $M_b(t)$  is an exponential decay function.

It is reasonable to expect sufficient medicine to be available within the body for a 24 hour period. Euler's approximations to the SCM in figure 3.1 show that values of  $\alpha$  that meet this requirement range for 0.15 to 0.07. These values give some indication of the levels for the kidney diffusion rate  $d_2$  in the two compartment model.

### 3.2 GI Diffusion

Simulation of the SCM in the previous section considers only elimination from the body. The natural next step in modelling strategy is to investigate the intake side by itself. This is another type of single compartment model that considers transfer from GI tract to bloodstream shown in figure 3.2.

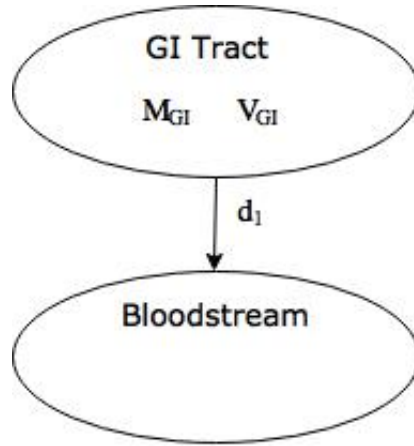


Figure 3.2: **GI diffusion only model (GID)**. Medicine diffuses from GI tract to bloodstream

The GI Diffusion only model (GID) sets the diffusion rate into the kidneys ( $d_2$ ) to zero. There is the further approximation that  $C_{GI}$  is constant. When a patient first takes a dose of medicine, there is no medicine in the bloodstream and  $M_{blood} = 0$ . The equation for the rate of change of  $M_{blood}$  is now:

$$\frac{dM_{blood}}{dt} = d_1 \left( C_{GI} - \frac{M_{blood}(t)}{V_{blood}} \right) \quad (3.1)$$

$$M_{blood}(0) = 0$$

### 3.2.1 Selecting GI concentration $C_{GI}$ and GI diffusion rate $d_1$

It is worth noting that the term  $(C_{GI} - \frac{M_b}{V_b})$  describes the concentration gradient between GI tract and bloodstream. Implementing Euler's method to approximate the solution to this equation and using the value for  $C_{GI}$  estimated in section 2.3.3 produces the plot in figure 3.3a. This curve fails to converge regardless of the value of  $d_1$ . Indeed, for  $d_1 = 0.5$ , the resultant amount of medicine exceeds the initial dose. For the other two values of  $d_1$ , the amount of medicine going into the bloodstream increases in a linear manner. An estimate of 100 mg for  $C_{GI}$  is not correct for this model. However, setting  $C_{GI}$  to 65 mg/l produces the plot in figure 3.3b.

From the plots for different  $d_1$ , a value of 1 for the GI to blood diffusion rate gives the expected exponential rise in amount of medicine. This value concurs with the Alexandria study [7].

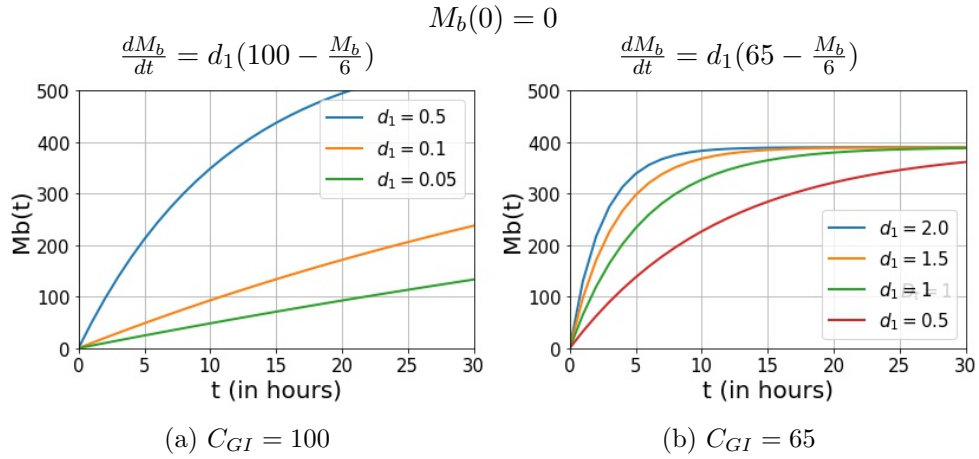


Figure 3.3: **GID Model and effect of  $C_{GI}$**  Comparing values for  $C_{GI}$  at different levels of GI diffusion rate  $d_1$ .

### 3.2.2 GID Model Stepsize $\Delta t$

The stepsize experiments take the highest value of  $d_1 = 2.0$ . If there are problems with convergence, they will occur when the slope of the curve is at its steepest (section A.2). Given that  $t_{max}$  is 3.9 hours, values of stepsize in the region of 1 hour seem appropriate.

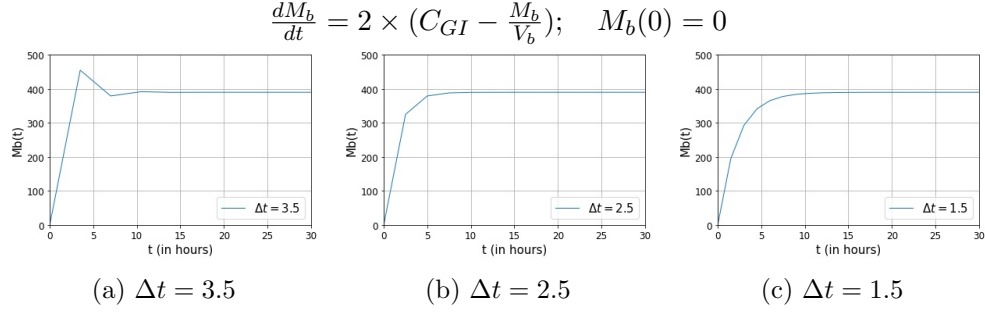


Figure 3.4: Effect of stepsize on convergence.

From figure 3.4c a stepsize of  $\Delta t = 1.5 \text{ h}^{-1}$  achieves convergence and results in a 'smooth' curve plot. Setting  $d_1 = 1.0$  gives similar results.

### 3.2.3 GID Estimation of Error

The GID model to investigate accuracy uses the value of  $d_1 = 1 \text{ h}^{-1}$  arrived at in section 3.2.1. Also, it uses the value of  $\Delta t = 1.5$  arrived at in section 3.2.2 and takes lower values to observe difference in the estimated curves for  $M_b(t)$ .

The steepest part of the curve is where potential problems with error may arise. Figure 3.5 displays this at the time line  $t = 6 \text{ h}$ . This plot truncates at  $t = 14 \text{ h}$  in order to highlight the effect of different values of  $\Delta t$ . As shown, it is only when  $\Delta t$  changes by an order of magnitude that the approximations show noticeable difference. This suggests that curve for  $\Delta t = 1.5 \text{ h}$  is an accurate estimate.

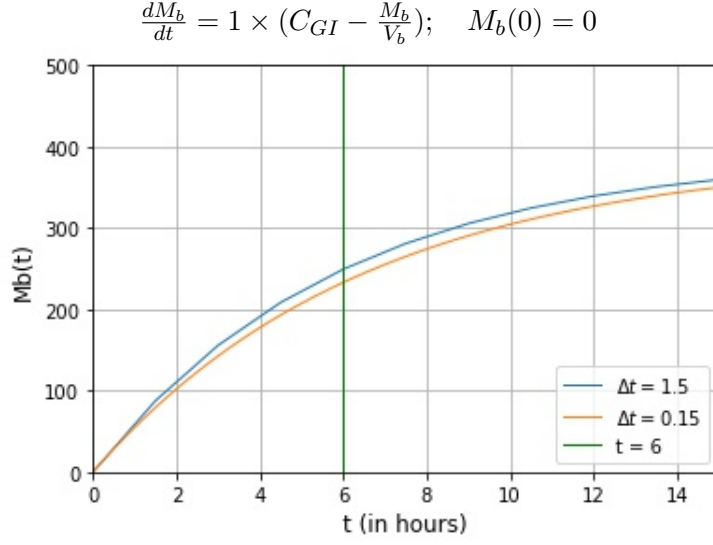


Figure 3.5: **Effect of stepsize on accuracy.** Stepsize in GID Model at steepest part of curve.

Nevertheless, it is worth following the procedure in [3] to calculate error. First it is necessary to decide on the level of accuracy for  $M_b(t)$ . The maximum value is 400 mg ( $D_e$ ). In non-critical medical studies, a false positive rate of 5

| $\Delta t$ | $M_b(6)$ | $w\Delta t - w2\Delta t$ |
|------------|----------|--------------------------|
| 3.0        | 292.5    |                          |
| 1.5        | 266.6    | 25.9                     |
| 0.75       | 256      | 10.6                     |
| 2          | 274.4    |                          |
| 1          | 259.4    | 15                       |

Table 3.1: **Error calculation** Approximate error from twice stepsize difference in GID Model

As table 3.1 shows, a stepsize of  $\Delta t = 1.5$  is outside the accuracy bound. A stepsize of  $\Delta t = 1$  does meet the requirement since the error here is 15.

### 3.3 Renal Diffusion

This section focuses on diffusion from the bloodstream to the kidneys, as shown in figure 3.6. This is another type of single compartment model and the same mathematical model as in section 3.1 applies, except that the analysis in this case takes account of blood volume  $V_b$ . The formula for the medicine

differential for this Renal Diffusion (RD) model now becomes:

$$\frac{dM_{blood}}{dt} = -d_2 \frac{M_{blood}(t)}{V_{blood}} \quad (3.2)$$

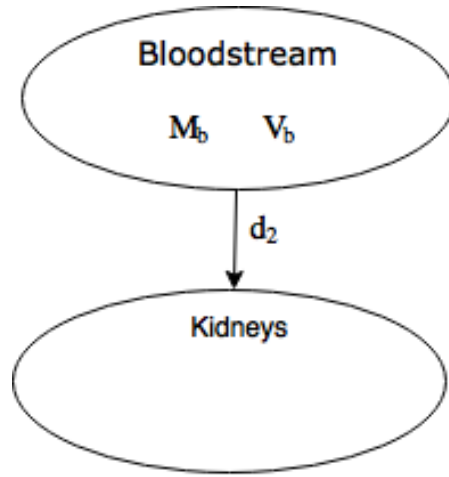


Figure 3.6: **Renal Diffusion Model** Diffusion from bloodstream to kidneys.

### 3.3.1 Selecting Renal Diffusion Rate $d_2$

In order to model the extreme case, the Euler calculation sets the initial value for  $M_b$  at twice the effective dose (800 mg). Taking the results from section 3.1 as a starting point, first consider values of  $d_2$  from the SCM model 3.1, which are in the range of 0.15 to 0.07. Euler approximation with these values shows that, even after 30 hours, there is still 400 mg in bloodstream. This suggests that higher values for  $d_2$  are required.

Simulation results for the value of the GI diffusion rate  $d_1$  from section 3.2.1 correlate with the Alexandria study [7]. This suggests using similar values for the renal diffusion rate  $d_2$  as the Alexandria study also. These range from 0.2 to 0.7.

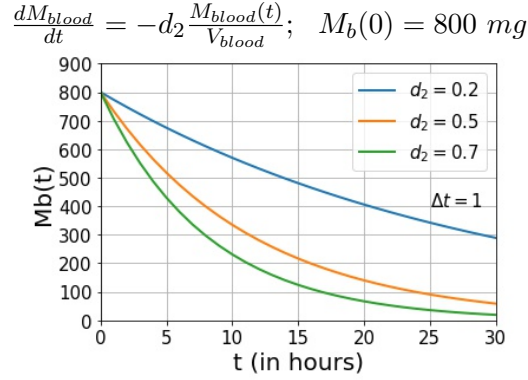


Figure 3.7: **Renal Diffusion** Diffusion from bloodstream to kidneys for values of  $d_2$ .

Figure 3.7 shows that, with  $d_2 = 0.7 \text{ h}^{-1}$ , the amount of medicine remaining in the blood stream goes below 100 mg after about 15 hours. With  $d_2 = 0.5 \text{ h}^{-1}$ , it takes 25 hours to reach the same level. The other value of  $d_2 = 0.2 \text{ h}^{-1}$  is too low to clear the medicine over the period of a day. So values of  $d_2$  at 0.5 to 0.7 are appropriate.

### 3.3.2 RD Model Stepsize $\Delta t$

From figure 3.7, the plot that has the greatest rate of change is  $d_2 = 0.7 \text{ h}^{-1}$ . Consider stepsizes in the vicinity of 1 h, which was the conclusion with the GID model in section 3.2.2. Figure 3.8a shows that, as with the GID Model, a stepsize of 1 hour results in a 'smooth' curve. When stepsize is 5 or greater, the curve is 'jagged' and the approximation of the tangent line for the differential is apparent.

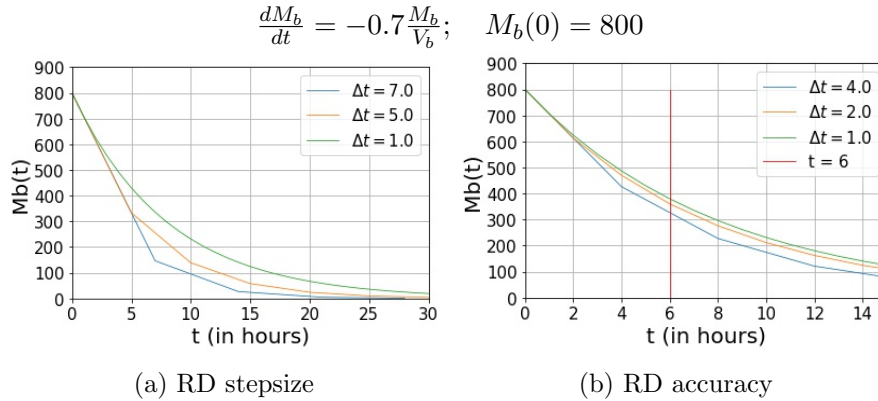


Figure 3.8: **Selecting stepsize and determining accuracy.** Effect of stepsize and level of accuracy with the RD Model

### 3.3.3 RD Model Accuracy

Figure 3.8a shows that the steepest part of the  $M_b(t)$  plots is near  $t = 6$  h. In addition, this value facilitates error calculation. In order better to display error, the plot in figure 3.8b truncates at  $t = 14$ .

| $\Delta t$ | $M_b(7)$ | $w\Delta t - w2\Delta t$ |
|------------|----------|--------------------------|
| 3.0        | 338.0    |                          |
| 1.5        | 370.6    | 32.6                     |
| 0.75       | 384.55   | 13.94                    |
| 2          | 360.50   |                          |
| 1          | 242.91   | 19.55                    |

Table 3.2: **Error calculation** Approximate error from twice stepsize difference in RD Model

Table 3.2 gives the error estimation at  $\Delta t = 1$  to be 19.55. This is just within the required level of accuracy of 20 mg.

## Chapter 4

# Two Compartment Model

Recall the formula for describing the change in quantity of medicine in the bloodstream:

$$\frac{dM_{blood}}{dt} = d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}}$$

Results for Single Compartment Model Analysis:

$$d_1 = 1.0 \text{ h}^{-1}$$

$$d_2 = 0.7 \text{ h}^{-1}$$

$$\Delta t = 1 \text{ h}$$

$$V_{GI} = 6 \text{ l h}$$

### 4.1 Constant GI concentration

Figure 4.1 shows the effect of a single dose of 500 mg of medicine and  $C_{GI}$  at the constant value of 65 mg/litre, as concluded in section 3.2.1.

$$\frac{dM_{blood}}{dt} = 1.0 \times \left(65 - \frac{M_{GI}(t)}{V_{GI}}\right) - d_2 \frac{M_{blood}(t)}{V_{blood}}; \quad M_b(0) = 0$$

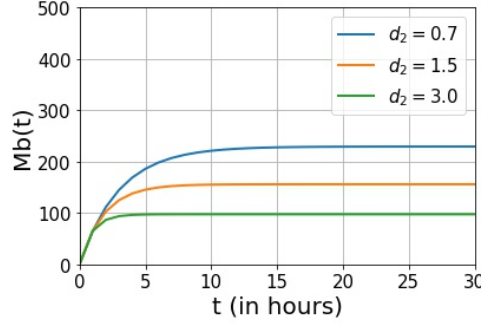


Figure 4.1: **TCM with constant  $C_{GI}$**  Two Compartment Model for  $M_{blood}(t)$  only and constant concentration of drug in GI tract.

#### 4.1.1 Validation

Figure 4.1 highlights the deficiency in the assumption of constant concentration of medicine in the GI tract. While it was useful in determining values for the diffusion rates, it does mean that there is a continuous supply of medicine available in the GI tract. Even setting  $d_2$  to three times  $d_1$ , the model does not produce the expected exponential decrease in  $M_b$ . What is happening is that the model reaches an equilibrium point that is unrealistic.

## 4.2 Full System Model - Single Dose

In order to study the confluence of diffusion into the bloodstream and elimination into the kidneys, it is necessary to take account of change of medicine in both GI tract and bloodstream simultaneously, as formulated in equations 2.1 and 2.2. Recall the equation for the medicine differential in the GI tract:

$$\frac{dM_{GI}}{dt} = -d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right)$$

So instead of  $C_{GI}$ , there is the term  $\frac{M_{GI}}{V_{GI}}$  where  $V_{GI}$  is given by [11] at 4 l and  $M_{GI}$  varies.

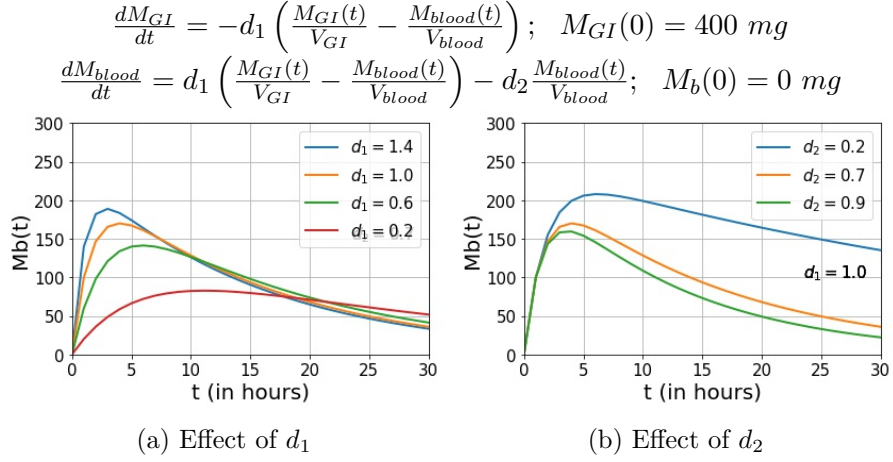


Figure 4.2: **Full System Model.** Effect of varying  $d_1$  and  $d_2$  with the Full System Model

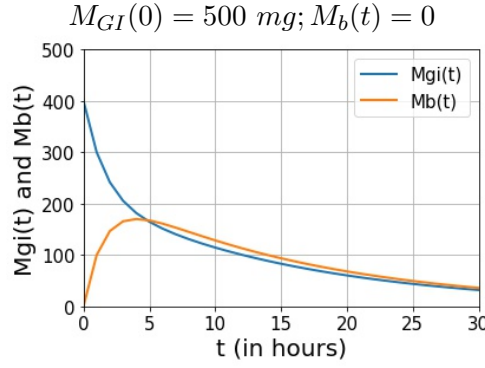


Figure 4.3: **TCM with  $M_{GI}$  and  $M_b$**  Two Compartment Model for amounts of medicine in both GI tract and bloodstream.

In figure both figures above, there is a single dose of 500 mg which gives an effective dose of 400 mg. Figure 4.3 shows the change in both  $M_{GI}$  and  $M_{blood}$ . This plot comes from the Euler calculation of simultaneous differential equations given in appendix C. It is worth noting that this algorithm produces the same result as vector functions that used by Keijzer [4]. This is because, while  $M_{GI}$  and  $M_b$  appear in both equations, there is no interaction term.

#### 4.2.1 Validation

Figure 4.2a shows the effect of different values of  $d_1$ . The high rate of  $1.4 \text{ h}^{-1}$  results in quick rise and drop in medicine level. The result is an initial boost that does not last long. This is not what doctors intend for patients. The

rate of  $0.6 \text{ h}^{-1}$  peaks at about  $M_b = 150 \text{ mg}$  and may not reach sufficient potency. With  $d_1 = 0.2 \text{ h}^{-1}$ , the amount of medicine does not go above 80 mg, which is inadequate for an administered dose of 500 mg. The rate of rate of  $d_1 = 1.0 \text{ h}^{-1}$  produces the expected behaviour of exponential rise and followed by exponential decline with a damping effect.

Figure 4.2b displays the effect of changing  $d_2$ . The low rate of  $0.2 \text{ h}^{-1}$  produces a situation where high levels of medicine (greater than 100 mg) persist in the body after 24 hours, which is unrealistic. The other two value of 0.7 and 0.9 results in the expected outcome.

It is interesting that this analysis for the changing amount of medicine in the two compartments of the body concludes with values for the diffusion rates, ( $d_1$  and  $d_2$ ), that are the same as the comparable variables ( $k_1$  and  $k_e$ ) in the Alexandria study [7] which modelled concentration instead of amount.

The above answers the research question in section 1.1.

### 4.3 Full System Model - Multiple Doses

It is necessary to consider the long term effect of repeated daily doses of medicine since this is the normal patient experience.

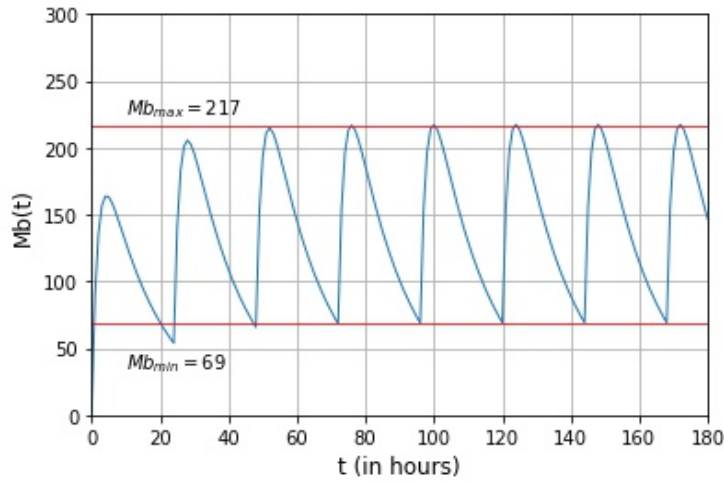


Figure 4.4: **Multiple Doses** Full System Model spanning 8 doses of medicine.

Figure 4.4 shows that the  $M_b$  reaches steady state after 5 doses. This concurs with the analysis in [1]. Given a dose of 500 mg, the amount that the patient receives is between 217 and 69 mg giving an average amount of 143 mg, thus

providing an approximate answer to the research question in section 1.1.

#### 4.3.1 Dose amount

To deal with the research question in section 1.1, the simulation here is with dose amounts of 1gm, 500mg and 250mg. So the effective dose ( $D_e$ ) is 0.8mg, 0.4mg and 200mg respectively.

| $D_e$ | $Mb_{max}$ | $Mb_{min}$ | $Mb_{avg}$ |
|-------|------------|------------|------------|
| 800   | 434        | 138        | 296        |
| 400   | 217        | 69         | 143        |
| 200   | 108        | 34         | 74         |

Table 4.1: **Dose amounts and averages** Actual amount of medicine in the bloodstream for different dose levels.

Table 4.1 suggests a linear relationship between Dose amount and level within the bloodstream, as measured by average values.

## Chapter 5

# Conclusions

The basis of this study are two differential equations for the change in medicine amount in the GI tract and change in the bloodstream. Inspection of these equations leads to expected behaviour of exponential rise followed by decay with a damping effect at higher values of medicine amount. There follows analysis of stability of a simplified model and the evaluation of quantities from known metrics. This section highlights concern about the volume of the contents of the GI tract.

The two main schema for studying the change in the amount of medicine in the body are the single compartment model and the two compartment model. A simplified model consists of a single compartment and holds the concentration of GI tract contents,  $C_{GI}$ , at a constant level and arrives at an appropriate value. This type of model allows focus on the GI diffusion rate  $d_1$  and the Renal diffusion rate  $d_2$  separately in order to arrive at appropriate values for these rates. In each case, there is analysis of the appropriate stepsize for Euler's method approximation and of the level of accuracy. In addition, there is validation of the value of  $C_{GI}$  appropriate to the mathematical model in this study.

The two compartment model accounts for the confluence of diffusion from GI tract to bloodstream and diffusion from bloodstream to kidneys. The first step is to consider the change in medicine amount in the bloodstream itself. It tests the assumption of constant GI concentration,  $C_{GI}$ , and points out a flaw.

The second stage models change in both GI tract and bloodstream simultaneously. Now it is possible to replace  $C_{GI}$  with the variables  $M_{GI}$  and  $V_{GI}$ . This results the behaviour expected from the mathematical model. Taking the case of a single dose, there is validation of suitable values for the diffusion rates,  $d_1$  and  $d_2$ .

The third step considers multiple doses, identifies when steady state is reached and calculates average amounts of medicine that the patient receives.

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# Appendix A

## Calculation

### A.1 Euler's Method

Euler's method is a numerical algorithm for approximating the solution to an unknown function when the rate of change of that function is known. So, for example, the rate of change can be increasing as for exponential growth or positive polynomial. Similarly, the rate of change decreases for exponential decay. Or the differential can transition from an increasing phase to an equilibrium point and into a decreasing phase, as is the case in figure 2.1. In many situations, it is not possible to derive an analytic solution to the differential equation. However, Euler's method does give an approximate solution. The following exposition for Euler's method is based on the analysis by Dawkins [9].

The general case of a first order, differential equation initial value problem has the form:

$$\frac{dy}{dt} = f(t, y(t)); \quad y(0) = y_0 \quad (\text{A.1})$$

where the initial value,  $y_0$ , is known. So the derivative of  $y(t)$  is a function of time and of  $y(t)$  itself.

Euler's method approximates an unknown function,  $y(t)$ , given two pieces of information:

1. the formula for the slope of the tangent to the curve  $y(t)$
2. the value of  $y(t)$  at some initial time

The first step approximates the solution to equation A.1 at some time  $t$  close to  $t_0$ . The differential of  $y(t)$  is the slope of the tangent to the curve  $y(t)$ , as

shown in figure A.1. At time  $t_0$ ,  $y(t)$  has the value  $y_0$ . The derivative at this point gives the slope of the tangent line:

$$\frac{dy}{dt} = f(t_0, y_0)$$

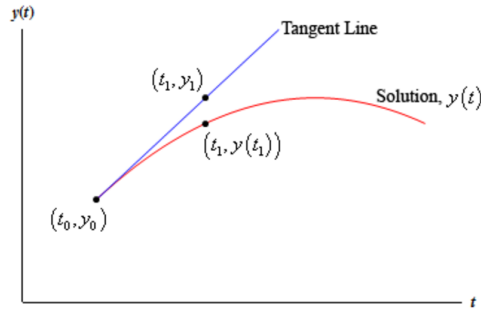


Figure A.1: **Solution and slope of tangent.** The differential to  $y(t)$  at  $t_0$  is the slope of the tangent at  $y_0$  (© 2003 - 2017 Paul Dawkins).

It is now possible to write the equation for the tangent line:

$$y = y_0 + f(t_0, y_0)(t - t_0) \quad (\text{A.2})$$

Consider advancing by time  $\Delta t$  to time  $t_1$ . The point on the tangent line is  $(t_1, y_1)$  and the equation for the tangent A.2 gives the value of  $y_1$ :

$$\begin{aligned} y_1 &= y_0 + f(t_0, y_0)(t_1 - t_0) \\ &= y_0 + f(t_0, y_0)\Delta t \end{aligned}$$

Euler's method makes the approximation that, if  $t_1 - t_0$  is sufficiently small, then the point on the tangent line is a good estimate for  $y(t_1)$ , the point on the curve of the solution function  $y(t)$ .

Euler's method proceeds in like manner by calculating the slope of the tangent at  $(t_1, y_1)$  by taking the value for the derivative,  $f(t_1, y_1)$ . This leads to an approximation for the next point at time  $t_2$ . Similarly, it is possible to approximate subsequent points on the solution by advancing in time by  $\Delta t$ . The result is a series of approximations  $A_i$  for the solution function, as shown in figure A.2.

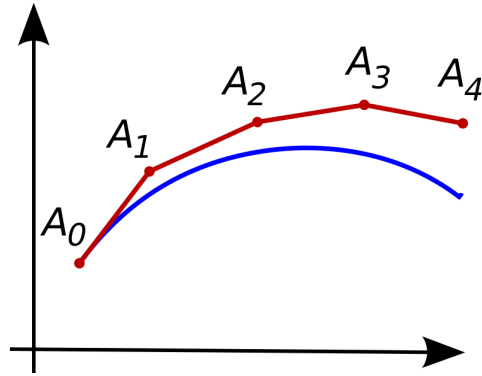


Figure A.2: **Euler's Method approximation.** The series of points  $A_1$  to  $A_4$  approximate the solution curve (O. Alexandrov).

The course notes [3] gives a derivation of Euler's method based on the integral over the time period  $\Delta t$ .

## A.2 Numerical Convergence

Choosing the appropriate step size is a critical factor in obtaining a good approximation to the solution curve. Figure A.3 shows that too large a step size leads to an erratic set of points that fail to converge (blue squares).

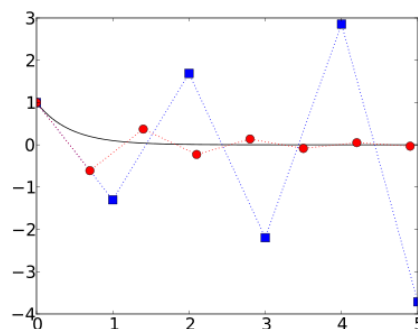


Figure A.3: **Instability of Euler's Method.** Euler's method approximations with step sizes  $\Delta t = 1$  (blue) and  $\Delta t = 0.5$  (red) [?].

### A.3 Accuracy

It is important to have some measure of how close Euler's approximation is to the exact solution of the function  $y(t)$ . Let the Euler approximation using step size  $\Delta t$  be  $w\Delta t$ . So the error for this value of step size is:

$$E\Delta t = y(t) - w\Delta t$$

The exact function is unknown. However, as described in [?] it is possible to estimate this error by the difference between the approximations at  $\Delta t$  and  $2\Delta t$ :

$$E\Delta t \approx w\Delta t - w2\Delta t$$

In practice, the problem specification tolerates a given amount of error. This guides selection of step size to a value sufficient to meet the accuracy requirement.

## Appendix B

# Derivations

### B.1 Derivation of SCM equation

Equation 1.2 is a first order, seperable differential equation and it is possible to solve analytically, given the initial condition that  $M = M_0$  at time  $t = 0$ :

$$\frac{dM}{dt} = -\alpha M(t)$$

$$\frac{1}{M(t)} \frac{dM}{dt} = -\alpha \quad < \text{divide by } M(t) >$$

$$\Rightarrow \int_{M_0}^{M(t)} \frac{1}{M(t)} dM = -\alpha \int_0^t dt \quad < \text{can separate change in time when integrate} >$$

$$\Rightarrow \ln|M(t)| \Big|_{M_0}^{M(t)} = -\alpha t \Big|_0^t \quad < \frac{d}{dx} \ln(x) = \frac{1}{x} >$$

$$\Rightarrow \ln|M(t)| - \ln|M_0| = -\alpha t$$

$$\Rightarrow M(t) = M_0 e^{-\alpha t}$$

## Appendix C

### Program Code

```
def calc_TCM(Mgi_init = 400, Mb_init = 0, d1=1.0, d2=0.7):
    """
    Compute approx to simultaneous differential equns using Euler's method

    Arguments
    -----
        Mgi_init: int, effective dose
        Mb_init:  int, initial amount of medicine in bloodstream
        d1: float, GI to bloodstream diffusion rate
        d2: float, bloodstream to kidneys diffusion rate

    Returns
    -----
        t_arr: numpy array, time values increasing by Dt
        Mgi_arr: numpy array, approx values of Mgi
        Mb_arr: numpy array, approx values of Mb

    """
    print "Solution for: "
    print "    dMgi/dt = - d1 * ( Mgi/Vgi - Mb/Vb ) "
    print "    dMb/dt = d1 * (Mgi/Vgi - Mb/6) - d2 * Mb/Vb " # in Python 2.7: use no
    print "    Mgi(0) = 400; Mb(0) = 0"

    # Initializations
    Dt = 1                                # stepsize
    t_init = 0                            # initial time
    t_end = 30                            # stopping time
    d1 = 1.0                              # diffusion rates 1/h
    d2 = 0.7
```

```

Vgi = 4
Vb = 6

n_steps = int(round((t_end-t_init)/Dt)) # total number of timesteps
t_arr = np.zeros(n_steps + 1)          # create an array of zeros for t
Mgi_arr = np.zeros(n_steps + 1)        # and for Mgi and Mb
Mb_arr = np.zeros(n_steps + 1)
t_arr[0] = t_init                       # add the initial values to arrays
Mgi_arr[0] = Mgi_init
Mb_arr[0] = Mb_init

# Euler's method

for i in range (1, n_steps + 1):
    Mgi = Mgi_arr[i-1]
    Mb = Mb_arr[i-1]
    t = t_arr[i-1]
    dMgidt = - d1 * (Mgi/Vgi - Mb/Vb)
    dMbdt = d1 * (Mgi/Vgi - Mb/Vb) - d2 * Mb/Vb
    Mgi_arr[i] = Mgi + Dt* dMgidt        # calculate the next time step
    Mb_arr[i] = Mb + Dt* dMbdt
    t_arr[i] = t + Dt                    # adding the new t-value to the list

return t_arr, Mgi_arr, Mb_arr

```